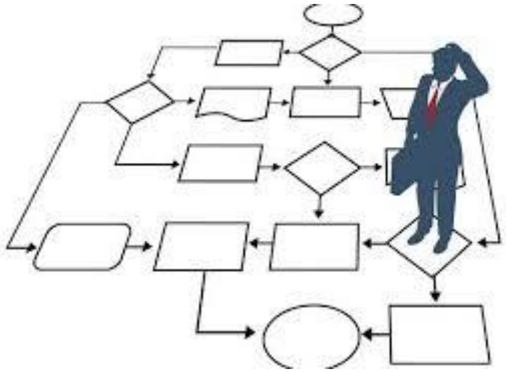
CSE346
BPM
Lect 4
“Process Analysis”



Dr. Islam El-Maddah

Once I’ve got a model, what’s next?

Analyze:

- Cycle time
- Capacity, resource utilization
- Cost
- QoS
- Risk
- …

Process Analysis – Typical Questions

What processes do we perform?

What value does each process generate?

What risks are embedded in this process?

What are our benchmarks and how close are we?

What change constraints exist (e.g. policies, IT, culture)?

What are the perceived and actual process problems and what is their impact?

Process Analysis – Typical Questions (cont.)

How much do existing processes cost?
How scalable and flexible are processes?
How do processes align with strategy?

When will changes be implemented (internal/environment)?

Where are the critical customer interfaces?
(See paper on “Analyzing and Improving Customer-Facing Processes” in the Readings list)

Why are the processes executed this way?

Who is accountable for these processes?

Business Process Analysis Techniques

- Qualitative Analysis
  - Strategy Maps / Balanced Scorecard Human Performance Analysis Issue Register
- Quantitative Analysis
  - Cycle Time Analysis
  - Capacity Analysis
  - Queuing Theory
  - Process Simulation

Issue Register

- Purpose: to categorise identified issues as part of as-is process modelling
- Usually a table with the following columns:
  - issue number
  - name
  - description
  - consequence
  - priority
  - type (IT / organ. / policy)
  - Impact: Qualitative vs. Quantitative
  - Possible solution

Cycle Time Analysis: Basic Concepts

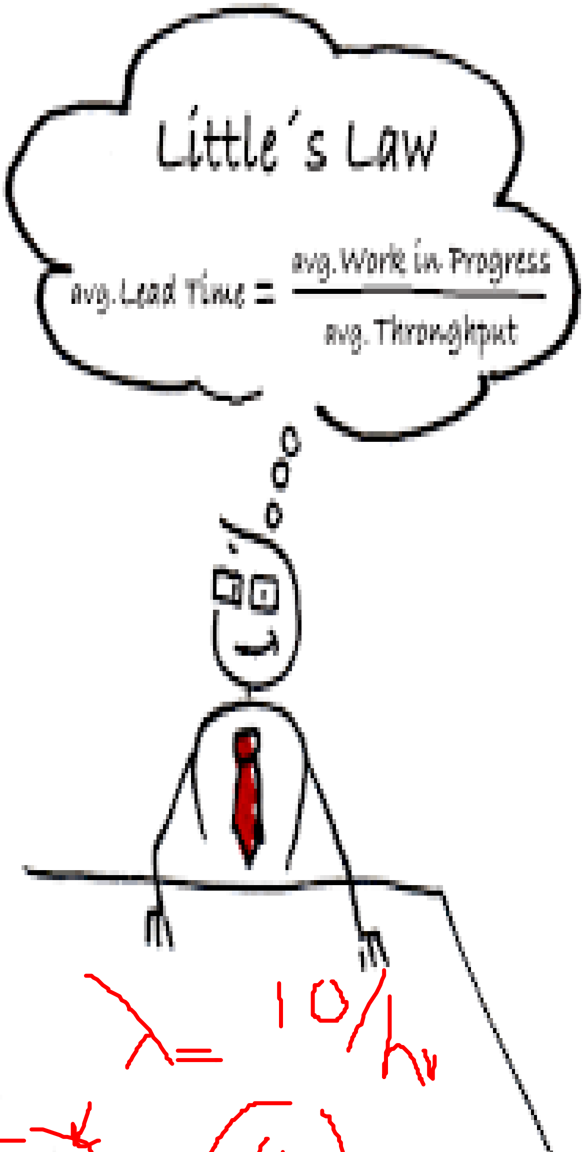
- Cycle time: Difference between a job's start time and end time
- Throughput: Number of jobs arriving per time unit
- Work-In-Progress: Average number of jobs that have entered the process but not yet left
- A long lasting trend in manufacturing has been to lower WIP by reducing batch sizes
  - The JIT philosophy
  - Forces reduction in set up times and set up costs

Cycle Time and WIP

- Little's Formula: General relationship between: WIP, the throughput (λ) and Cycle time (CT)

***Little's Formula*: $\text{WIP} = \lambda \cdot \text{CT}$**

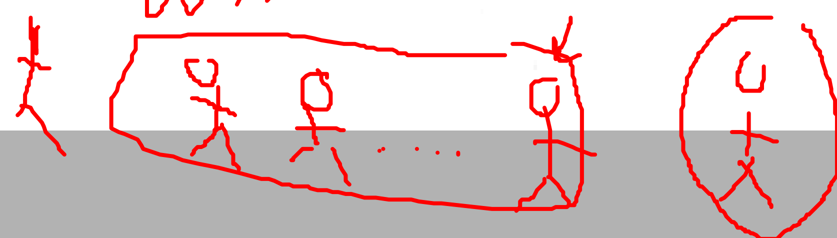
- Implications, everything else equal
  - Shorter cycle time $\Leftrightarrow$ lower WIP
  - If λ increases $\Rightarrow$ to keep WIP at current levels CT must be reduced



CT = 1 hr

WIP = 10

$\lambda = 10/\text{hr}$



Exercise 1

WIP = 30

A fast-food restaurant receives on average **1200** customers per day (between 10:00 and 22:00). During peak times (12:00-15:00 and 18:00-21:00), the restaurant receives around **900** customers in total, and **90** customers can be found in the restaurant (on average) at a given point in time. At non-peak times, the restaurant receives **300** customers in total, and **30** customers can be found in the restaurant (on average) at a given point in time.

1. What is the average time that a customer spends in the restaurant during peak times?

   $\frac{900}{6} = 150 = 90 / CT \Rightarrow 36$ min

2. What is the average time that a customer spends in the restaurant during non-peak times?

   $\frac{300}{6} = 50 = 30/C$ 36 min

Exercise 1 (continued)

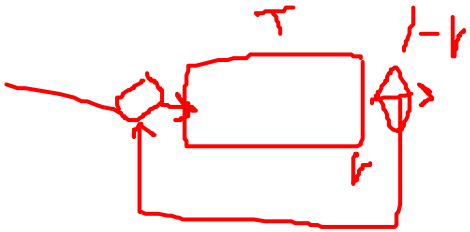
3. The restaurant plans to launch a marketing campaign to attract more customers. However, the restaurant's capacity is limited and becomes too full during peak times. What can the restaurant do to address this issue without investing in extending its building?

Cycle Time Analysis

- *Cycle time analysis*: the task of calculating the ***average*** cycle time for an entire process or process segment
  - Assumes that the average activity times for all involved activities are available (activity time = waiting time + processing time)
- In the simplest case a process consists of a sequence of activities on a single path
  - The average cycle time is the sum of the average activity times
- … but in general we must be able to account for
  - Rework
  - Multiple paths
  - Parallel activities

Rework

- Many processes include control or inspection points where if the job does not meet certain standard, it is sent back for rework
  - The rework will affect the average cycle time



- Definitions
  - T = sum of activity times in the rework loop
  - r = percentage of jobs requiring rework (rejection rate)
- Assuming a job is never reworked more than once

CT = (1+r)T

90T + .10 x 2T

1.1T

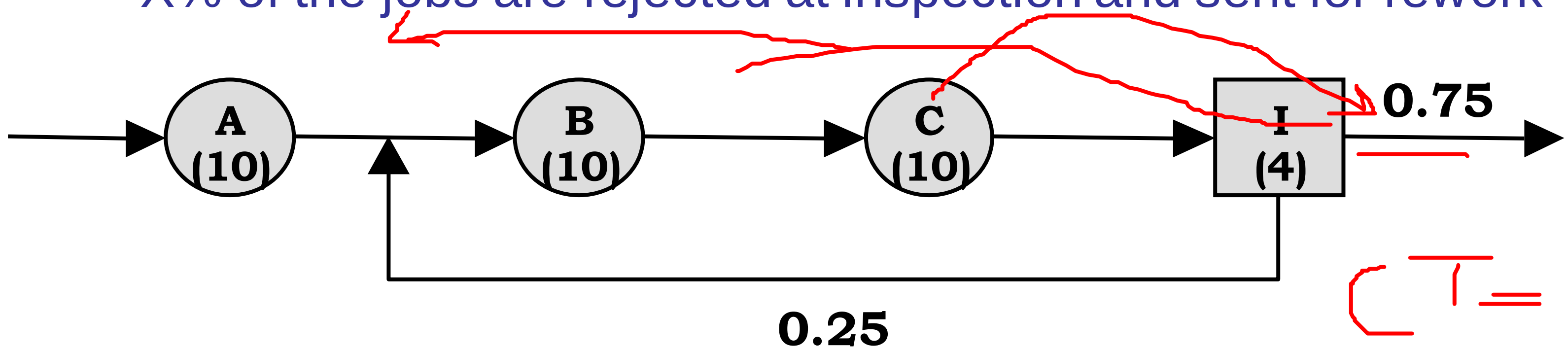
- Assuming a reworked job is no different than a regular job

CT = T/(1-r)

$\frac{T}{1-r}$ = 1.1111 T

Example – Rework effects on the average cycle time

- Consider a process consisting of
  - Three activities, A, B & C taking on average 10 min. each
  - One inspection activity (I) taking 4 minutes to complete.
  - X% of the jobs are rejected at inspection and sent for rework



$CT = 34$

➢ ***What is the average cycle time?***

a) If no jobs are rejected and sent for rework.

$CT = 10 + 24(1.25)$
$= 40$

b) If 25% of the jobs need rework but never more than once.

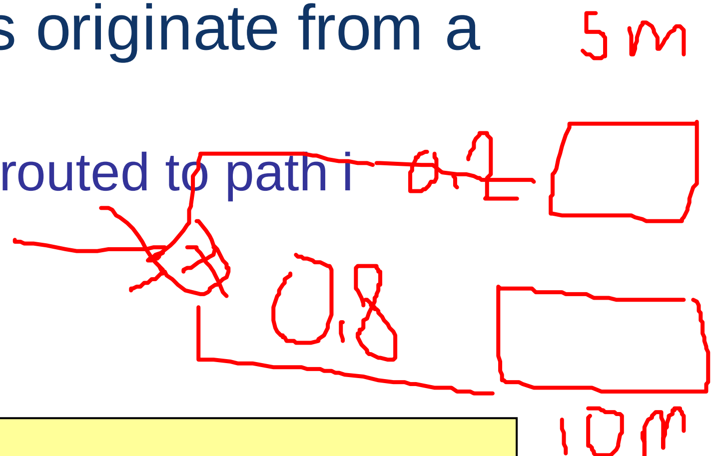
c) If 25% of the jobs need rework but reworked jobs are no different in quality than ordinary jobs.

$CT = 10 + \frac{24}{1-.25} = 10 + \frac{8}{.25} = 42$

Multiple Paths

- It is common that there are alternative routes through the process
  - For example: jobs can be split in “fast track”and normal jobs

- Assume that m different paths originate from a decision point
  - p_i = The probability that a job is routed to path i
  - T_i = The time to go down path i



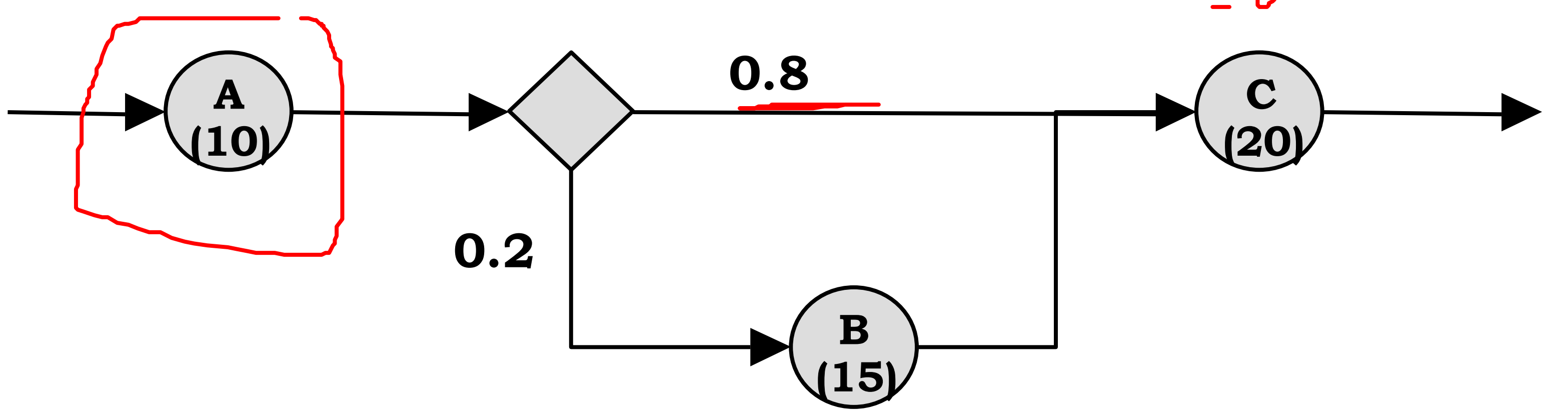
1 2 3

$$CT = p_1T_1+p_2T_2+\ldots+p_mT_m= \sum_{i=1}^{m} p_i T_i$$

Example – Processes with Multiple Paths

- **Consider a process segment consisting of 3 activities A, B & C with activity times 10,15 & 20 minutes respectively**
- **On average 20% of the jobs are routed via B and 80% go straight to activity C.**

$10 + 0.2 \times 15 + 20$



➢ ***What is the average cycle time?***

Processes with Parallel Activities

- If two activities related to the same job are done in parallel the contribution to the cycle time for the job is the maximum of the two activity times.
- Assuming
  - M process segments in parallel
  - T_i = Average process time for process segment i to be completed

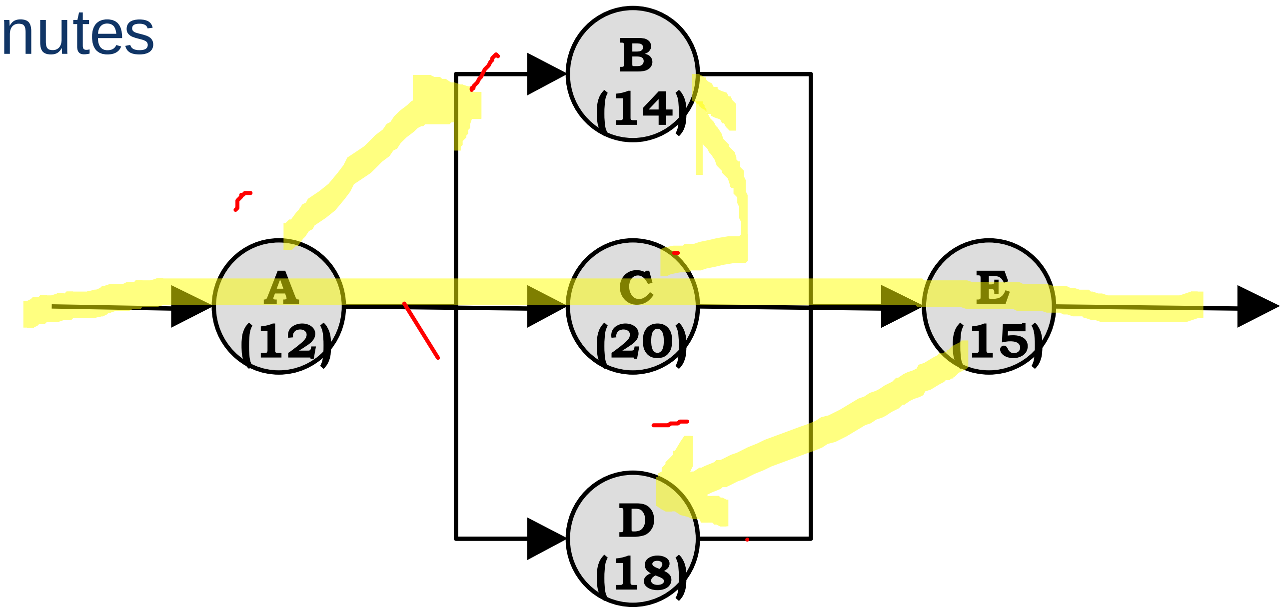
$$CT_{parallel} = Max\{T_1, T_2, \ldots, T_M\}$$

Example – Cycle Time Analysis of Parallel Activities

- Consider a process segment with 5 activities A, B, C, D & E with average activity times: 12, 14, 20, 18 & 15 minutes

WIP

$\lambda = 10/hr$



- ***What is the average cycle time for the process segment?***

Cycle Time Efficiency

- Measured as the percentage of the total cycle time spent on value adding activities.

$$\textbf{Cycle Time Efficiency} = \frac{\text{Theoretical Cycle Time}}{\text{CT}} \qquad \frac{CT}{TCT}$$

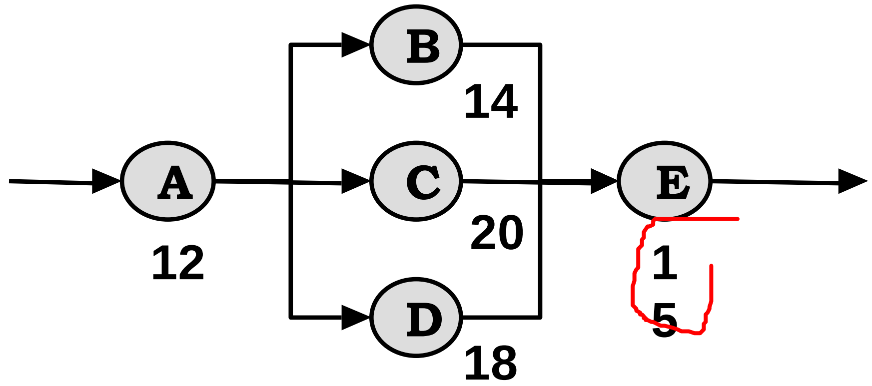
- Theoretical Cycle Time = the cycle time which we would have if only value adding activities were performed

Cycle time Reduction

- Cycle time analysis provides valuable information about process performance
  - Helps identify problems
  - Increases process understanding
  - Useful for assessing the effect of design changes
- Ways of reducing cycle times through process redesign
  1. Eliminate activities
  2. Reduce waiting and processing time
  3. Eliminate rework or reducing
  4. Perform activities in parallel
  5. Move processing time to activities not on the critical path

Example – Critical Activity Reduction

- Consider a process with three sequences or paths



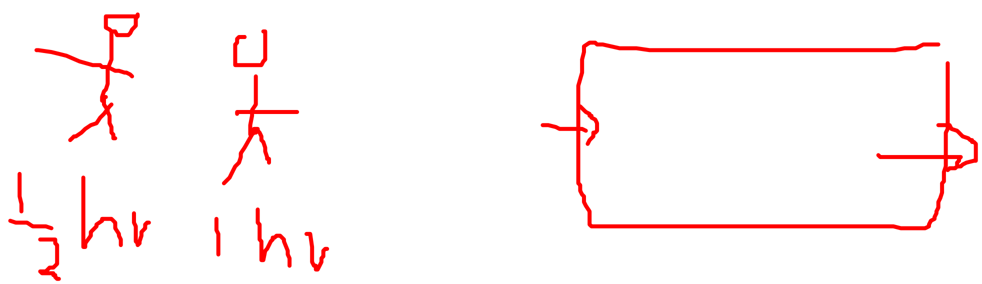
Sequence (Path)	Time required (minutes)
1. A→B →E	12+14+15 = 41
2. A→C →E	12+20+15 = 47 = ***CT***
3. A →D →E	12+18+15 = 45

Critical path

⇒ ***By moving 2 minutes of activity time from path 2 to path 1 the cycle time is reduced by 2 minutes to CT=45 minutes***

Limitations of Cycle Time/Capacity Analysis

- Cycle time analysis and capacity do not consider waiting times due to resource contention
- Cycle time analysis cannot be used to measure other important aspects such as cost
- Process simulation addresses these limitations



Process Simulation

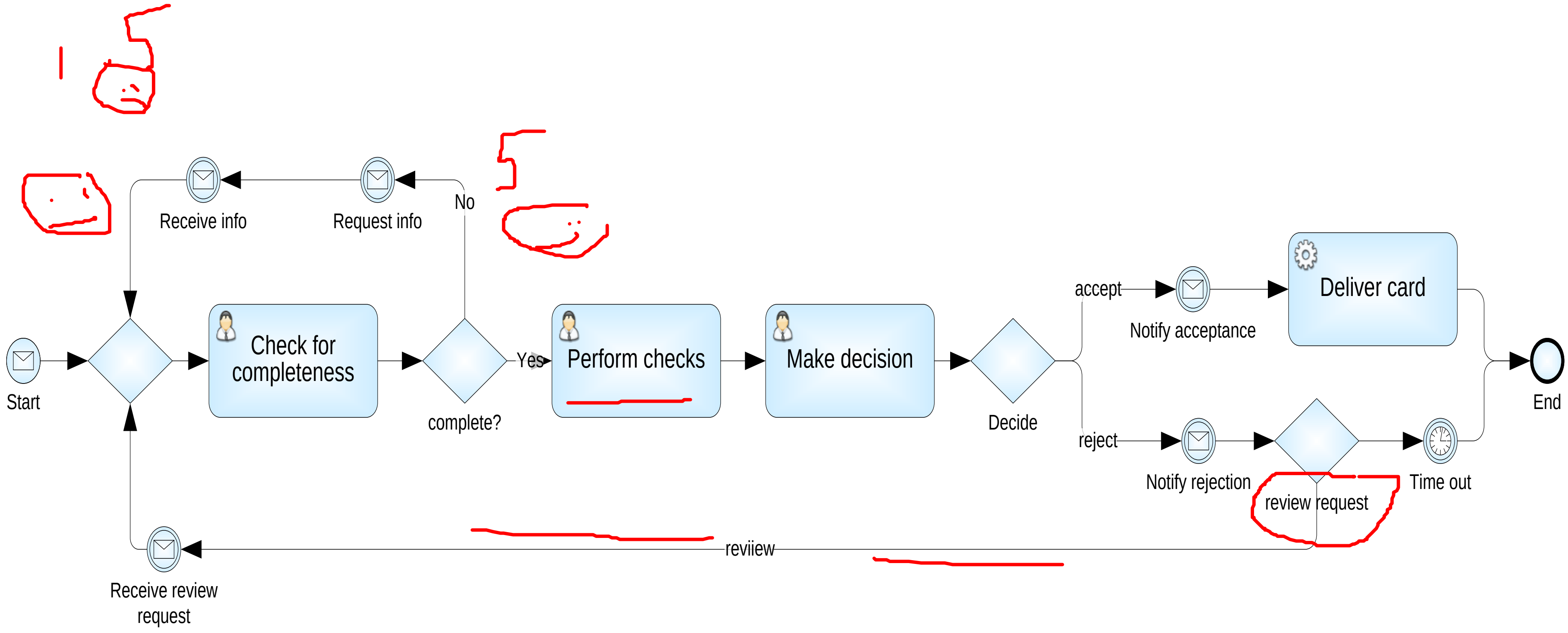
- Process simulation = run a large number of process instances, gather data (cost, duration, resource usage) and calculate statistics from the output
- Basic steps in evaluating a process model with simulation
    1. Building the simulation model
    2. Running the simulation
    3. Analyzing the simulation results (performance measure)
    4. Evaluation of alternative scenarios

KPI

Elements of a simulation model

- The process model including:
  - Activities, control-flow relations (flows, gateways)
  - Resources and resource pools (i.e. roles)
- *Resource requirements*: mapping between activities and *resource pools*
- Processing times (per activity, or per activity-resource pair)
- Costs (per activity, or per activity-resource pair)
- Arrival rate (also called: token creation)
- Conditional branching probabilities (XOR gateways)

Simulation Example – BPMN model



Resource Pools (Roles)

- Two options to define resource pools
  - Define individual resources of type clerk
  - Or assign a number of “anonymous” resources all with the same cost
- E.g.
  - 3 anonymous clerks with cost of € 10 per hour, 8 hours per day
  - 2 individually named clerks
    - Jim: € 12, 4 hours per day
    - Mike: € 14, 8 hours per day
  - 1 manager John at € 20 per hour, 8 hours per day

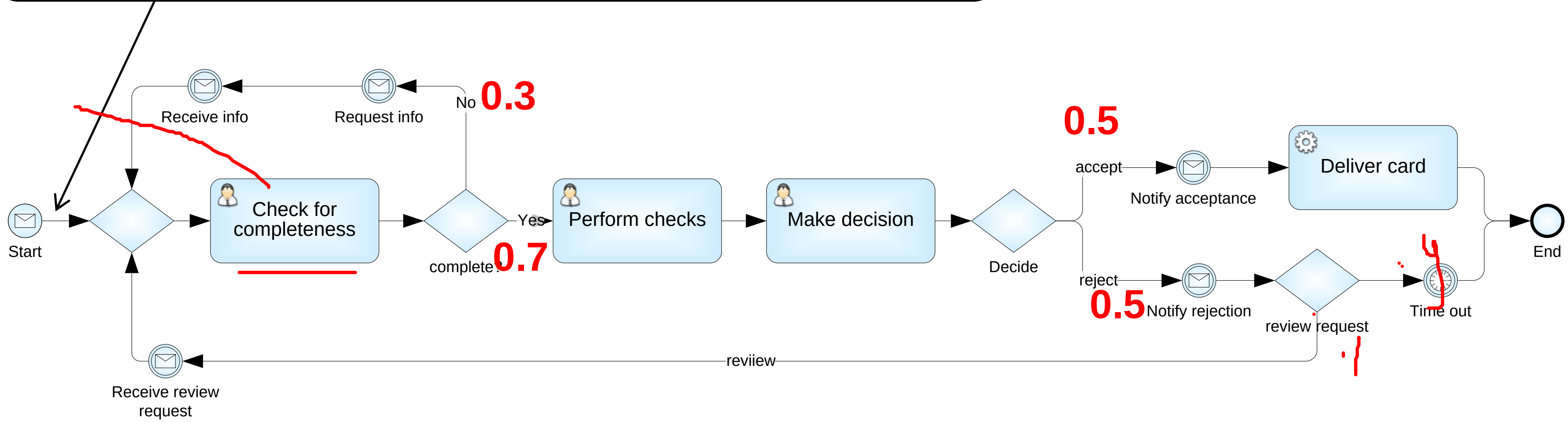
Resource pools and execution times

Task	Role	Execution Time Normal distribution: mean and std deviation	
Receive application	system	0	0
Check completeness	Clerk	30 mins	10 mins
Perform checks	Clerk	2 hours	1 hour
Request info	system	1 min	0
Receive info (Event)	system	48 hours	24 hours
Make decision	Manager	1 hour	30 mins
Notify rejection	system	1 min	0
Time out (Time)	system	72 hours	0
Receive review request (Event)	system	48 hours	12 hours
Notify acceptance	system	1 min	0
Deliver Credit card	system	1 hour	0

Alternative: assign execution times to the tasks only (like in cycle time analysis)

Arrival rate and branching probabilities

10 applications per hour (one at a time)
Poisson arrival process (negative exponential)



Alternative: instead of branching probabilities one can assign "conditional expressions" to the branches based on input data

Simulation output: KPIs

Simulation output: detailed logs

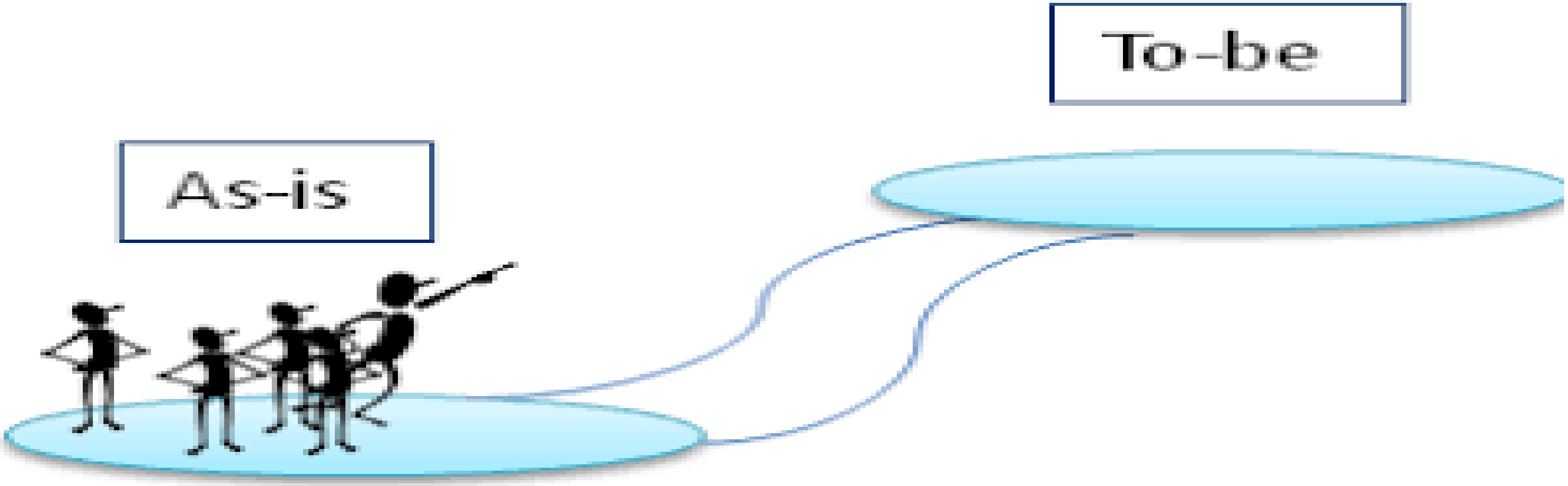
Process Instance	# Activities	Start	End	Cycle Time	Cycle Time (s)	Total Time
6	5	4/06/2007 13:00	4/06/2007 16:26	03:26:44	12403.586	03:26:44
11	5	4/06/2007 18:00	5/06/2007 12:14	18:14:56	65695.612	18:14:56
13	5	4/06/2007 20:00	5/06/2007 13:14	17:14:56	62095.612	17:14:56
16	5	4/06/2007 23:00	5/06/2007 15:06	16:06:29	57989.23	16:06:29
22	5	5/06/2007 5:00	6/06/2007 10:01	29:01:39	104498.797	29:01:39
27	8	5/06/2007 10:00	6/06/2007 12:33	26:33:21	95600.649	26:33:21

Process Instance	Activity ID	Activity Name	Activity Type	Resource	Start	End
	6aed54717-f044-4da1-b543-82a660809ecb	Check for completeness	Task	Manager	4/06/2007 13:00	4/06/2007 13:53
	6a270f5c6-7e16-42c1-bfc4-dd10ce8dc835	Perform checks	Task	Clerk	4/06/2007 13:53	4/06/2007 15:25
	677511d7c-1eda-40ea-ac7d-886fa03de15b	Make decision	Task	Manager	4/06/2007 15:25	4/06/2007 15:26
	6099a64eb-1865-4888-86e6-e7de36d348c2	Notify acceptance	IntermediateEvent	(none)	4/06/2007 15:26	4/06/2007 15:26
	60a72cf69-5425-4f31-8c7e-6d093429ab04	Deliver card	Task	System	4/06/2007 15:26	4/06/2007 16:26
	7aed54717-f044-4da1-b543-82a660809ecb	Check for completeness	Task	Manager	4/06/2007 14:00	4/06/2007 14:31
	7a270f5c6-7e16-42c1-bfc4-dd10ce8dc835	Perform checks	Task	Clerk	4/06/2007 14:31	5/06/2007 8:30

Simulation Tutorial

- http://developer.tibco.com/business_studio/getting-started.jsp
  - Business Studio Training Lab

Process Re-Design: From "as is" to "to be"



Re-Design Criteria

A process design is evaluated on the basis of four key issues:

- time
- quality
- costs
- flexibility

Often there is a trade-off!

Design criterion 1: Time

- Throughput time (see earlier), including
  - service time (including set-up)
  - transport time (can often be reduced)
  - waiting time
    - sharing of resources (limited capacity)
    - external communication (trigger time)
- Several ways to improve time properties:
  - Improve average
  - Improve variance
  - Increase ability to meet due dates
  - Increase perception of wait time

Design criterion 2: Quality

- External: satisfaction of the customer
  - Product: product meets specification/expectation.
  - Process: the way the product is delivered (service level)
- Internal: conditions of work
  - challenging
  - varying
  - controlling

There is often a positive correlation between external and internal quality.

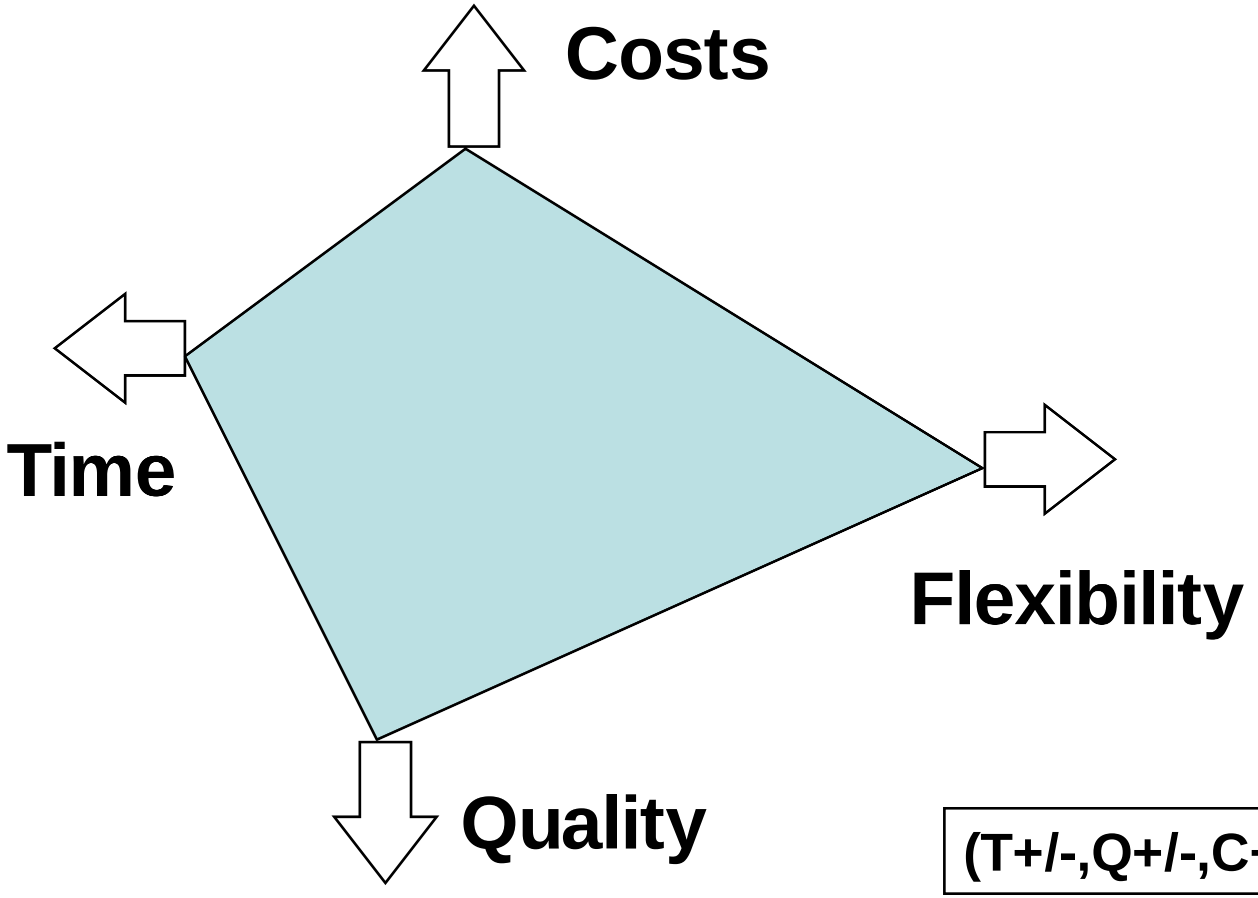
Design criterion 3: Cost

- Type of costs
  - fixed or variable
  - per time unit, per use (consumable resources)
  - processing, management, or support.
  - human, system (hardware/software), or external,

Design Criterion 4: Flexibility

- Ability to react to changes.
- Flexibility of
  - resources (ability to execute many tasks/new tasks)
  - process (ability to handle various cases and changing workloads)
  - management (ability to change rules/allocation)
  - organization (ability to change the structure and responsiveness to demands of market or business partners

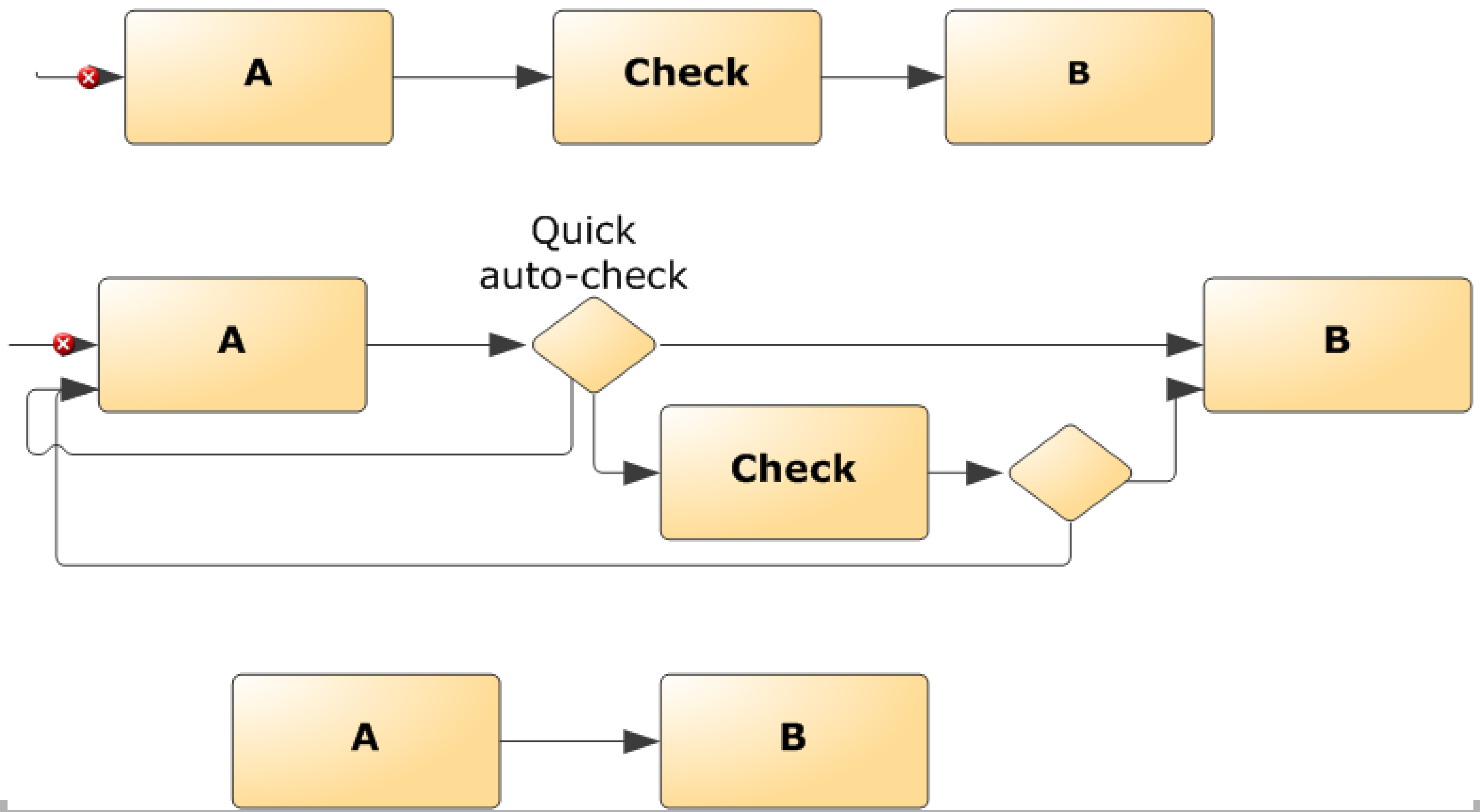
Trade-off



(T+/-,Q+/-,C+/-,F+/-)

(1) Check the necessity of each task

- Sometimes "checks" may be skipped: trade-off between the cost of the check and the cost of not doing the check.

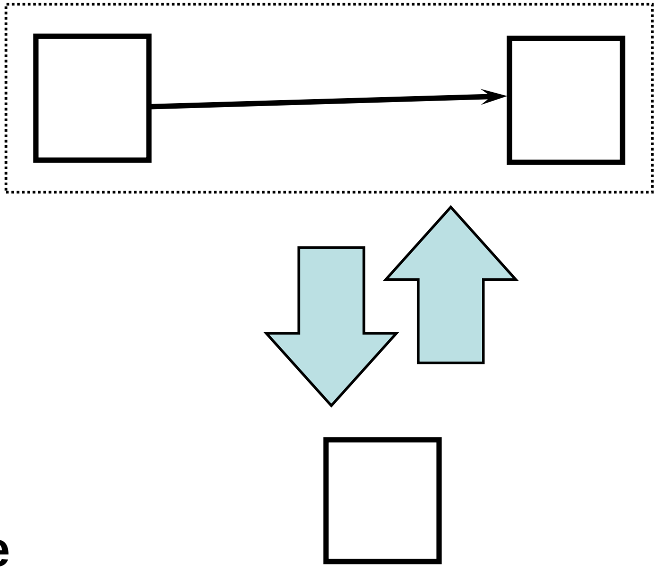


(T+,Q-,C+/-)

(1) Check the necessity of each task (cont.)

- Other tasks to consider for elimination:
  - Print
  - Copy
  - Archive
  - Store
  - More generally: non-value adding activities
- Task elimination can be achieved by delegating authority, e.g.
  - No need for approval if amount less than Y
  - Employees have budget for small expenses
  - Employees keep track of their own holidays, no authorization, just notification

(2) Re-consider the size of each task: merge or split



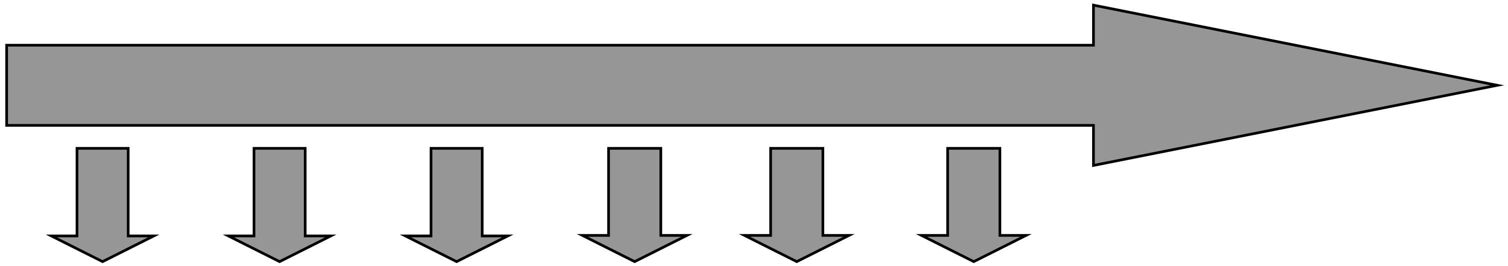
Pros: less work to commit, allows for specialization.
Cons: setup time, fragmentation, less commitment.

Pros: setup reduction, no fragmentation, less transportation time, more commitment
Cons: more work to commit, one person needs to be qualified for both parts.

Sometimes, splitting can be an opportunity to enable partial self-service, e.g. Decouple scanning and payment in a supermarket

(T+,F-)

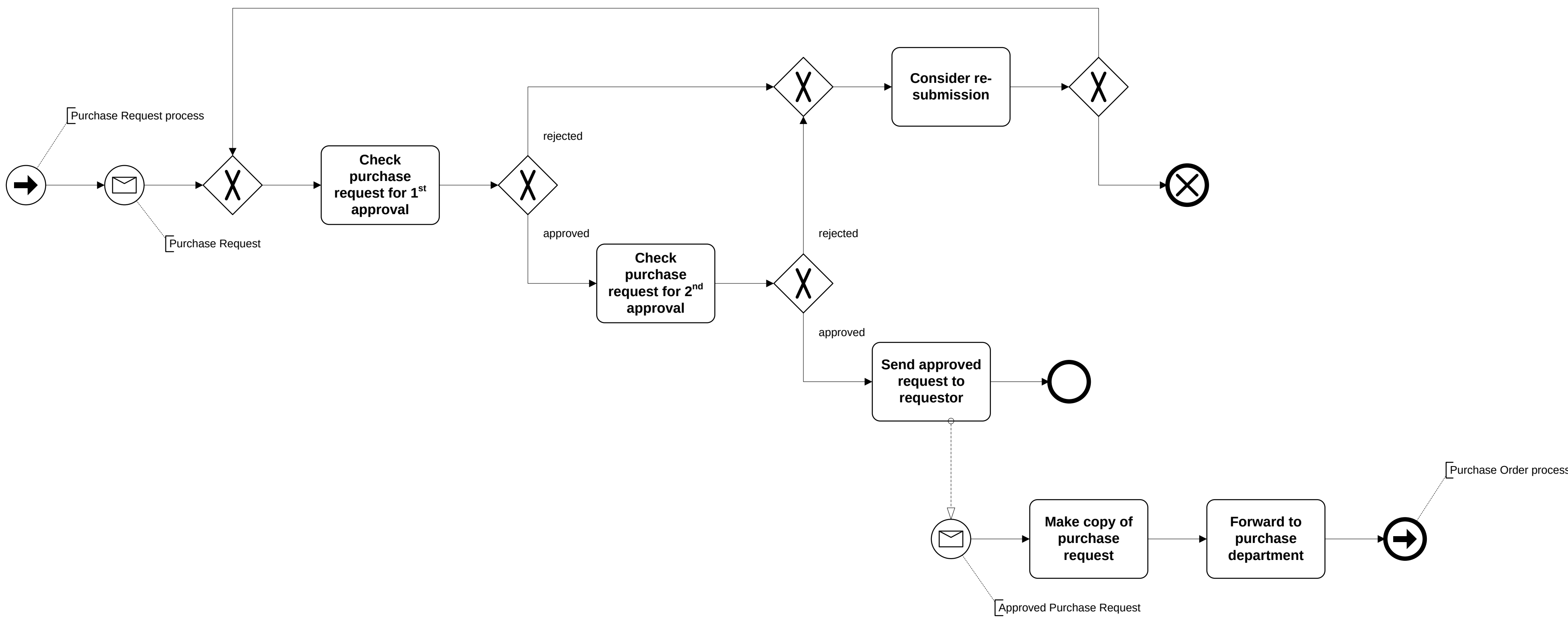
(3) Order tasks based on cost/effect



- Consider the class of “knock-out processes”, e.g., hiring people, handling claims, etc.
- Execute highly selective tasks first.
- Postpone expensive tasks until the end.
- In other words: order the tasks using the ratio “costs/effect”.

(T+,C-)

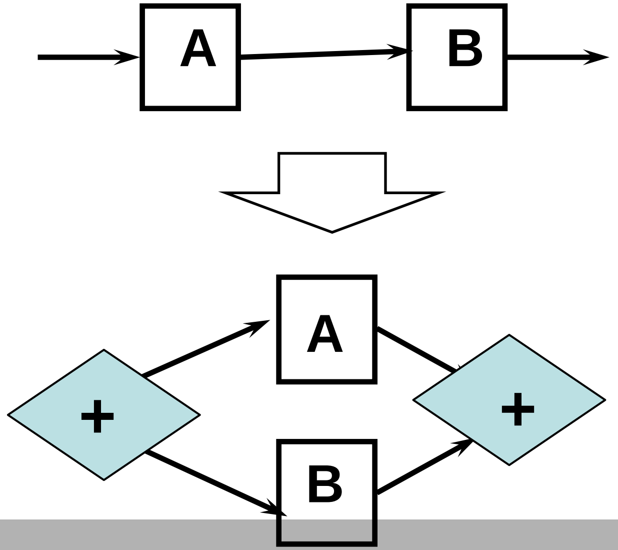
Example



Purchase Request process
Purchase Request
Check purchase request for 1st approval
rejected
approved
Check purchase request for 2nd approval
rejected
approved
Consider re-submission
Send approved request to requestor
Approved Purchase Request
Make copy of purchase request
Forward to purchase department
Purchase Order process

(4) Introduce parallelism

- More parallelism leads to improved performance: reduction of waiting times and better use of capacity.
- Two types of parallelism: semi and real parallelism.
- IT infrastructures which allow for the sharing of data and work enable parallelism.



(T++)

(5) Generic process vs. multiple versions

- Process customization
  - Differentiate by customer classes, geographical locations, time periods (winter, summer), …
  - Different activities, different resource pools,
- Process standardization
  - All cases treated equally (as much as possible)
  - Resources are pooled together

F+/-, C+/-

(6) Generic task vs. multiple specialized tasks

- Similar considerations.
- Specialization may lead to:
  - the possibility to improve the allocation of resources
  - more support when executing the task
  - less flexibility
  - a more complex process
  - monotonicity

(T+,F-)

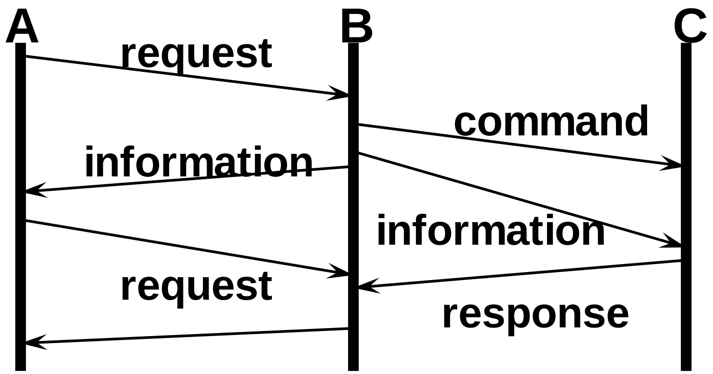
(7) Improve allocation of resources

- Use resources as if they are in one room: avoid one group of people overloaded and another (similar) group waiting for work.
- Let people do work that the are good at. However, avoid inflexibility as a result of specialization!
- Stimulate resources to build routine.
- When allocating work to resources, consider the flexibility in the near future.
- Avoid setups as much as possible. There are two kinds of setups: (1) case setups and (2) task setups.

(T+,Q-)

(8) Improve communication structure

- Reduce the number of messages to be exchanged between the process and the environment.
- Try to automate the handling of messages (send/receive).
- Avoid communication errors (EDI, XML, Web services)
- If possible, use asynchronous instead of synchronous communication.



(T+,Q+,C+/-,F-)

(9) Investigate IT-driven improvements

- Data sharing (Intranets, ERPs)
  - Increase availability of (subject to security/privacy) information to improve decisions or visibility
  - Avoid duplicate data entry, paper copies
- Use network technology to:
  - Increase communication speed: e-mail, SMS
  - Enable self-service (e.g. online forms)
  - Replace materials flow with information flow
- Tracking: RFID, GPS tracking
- Automation of tasks, automated support for tasks
- First re-design, then automate!

(T+,Q+/-,C+/-,F-)

(10) Appoint process/case managers

- A process manager monitors a process to see whether there are bottlenecks, capacity problems and delayed cases. Management instruments: motivating the people involved in the process and control parameters.
- Case managers are assigned to a case. They are responsible and execute as many tasks as possible for the case. Benefits:
  - commitment
  - reduction of setup time
  - one contact person

(Q+)

Homework 2 (10 points)

Continuation of homework 1

1. Write an issue register for the procurement process
2. Re-design your “as is” process model
   - → “to-be” process model

- Deliverables: Issue register and to-be process diagrams + oral presentation next session
- Can be done in groups of up to 4

Additional Activities

- In preparation for the project
  - Install TIBCO if you haven’t
  - Complete the process modeling and simulation tutorials (see previous slide on this topic)

Readings

(See course web page for links)

- P. Harmon. Analyzing Activities *BPTrends Advisor*, April 2003.
- P. Harmon. Analyzing and Improving Customer-Facing Processes *BPTrends Advisor*, December 2003.